

APARNA AGRAWAL

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AparnaAgrawal02



EDUCATION

International Institute of Information Technology, Hyderabad

B.Tech and MS in Computer Science and Engineering

2021 - 2025

CGPA: 7.59/10

PUBLICATION

“Gloss SLQA: A Gloss-centric Approach to Sign Language Question Answering” (Under review, PR Journal)

Aparna Agrawal*, Seshadri Mazumder*, C.V. Jawahar, Vinay P. Namboodiri

“Towards Scalable Sign Production: Leveraging Co-Articulated Gloss Dictionary for Fluid Sign Synthesis ” (ICVJPI 2025)

Aparna Agrawal*, Seshadri Mazumder, C.V. Jawahar, Vinay P. Namboodiri

“SignWeave-3D: Weaving Isolated Sign Units into Continuous Co-Articulated Motion” (Under submission at CVPR 2026)

Aparna Agrawal*, Seshadri Mazumder, C.V. Jawahar, Vinay P. Namboodiri

“An Approach for Speech Enhancement in Low SNR Environments using Granular Speaker Embedding to Enhance ASR” (CODSCOMADS 2024)

Jayasree Saha*, Rudrabha Mukhopadhyay, **Aparna Agrawal**, Surabhi Jain, C. V. Jawahar

RESEARCH EXPERIENCE

Centre for Visual Information Technology (CVIT), IIITH

Undergraduate Student Researcher

2022 - Present

Prof. C.V. Jawahar & Prof. Vinay Namboodiri

• 2D- Sign Language Production:

- * Developed an interpolation-based ISL sign production framework leveraging co-articulated gloss dictionaries and text-to-motion models, supported by a curated 1K-sentence gloss-annotated dataset.

• 3D co-articulation prediction using self-supervised text-to-sign motion pipeline :

- * Developed SignWeave-3D, a hybrid diffusion-MLM approach that bridges isolated sign dictionaries and continuous sign language generation, significantly improving co-articulation, temporal coherence, and out-of-vocabulary robustness.

• Sign Language Question Answering:

- * Developed novel pipeline combining sign language recognition, LLMs, and 3D Gaussian splatting.
- * Leveraged LLM to generate domain-specific vocab-constrained QnA pairs.
- * Improved sign language production.

• Speech Enhancement Research:

- * Developed novel granular speaker embedding techniques to improve ASR performance in challenging low SNR environments.

• Multilingual Content Accessibility:

- * Collaborated in enhancement of OpenShot video editor for conversion of academic lectures in different languages.
- * Integrated pipeline for seamless video conversion with audio translation, lip synchronization, and in-place PowerPoint language translation.

• Speech Separation:

- * Successfully demonstrated that generated lips could be used to separate speech from mixed audio.

• CVPR Reviewer:

- * Reviewed paper related to sign-language generation for CVPR 2025.

INDUSTRIAL EXPERIENCE

Goldman Sachs

Engineering Analyst

July - present
Bengaluru, India

- NLP Team.
- Real-time Conversation parsing.

Goldman Sachs

Software Development Intern

May - July 2024
Bengaluru, India

- Developed a proof-of-concept for recommendation system.
- Built a RESTful recommendation service.

Huddl.ai

AI Intern

June - July 2023
Hyderabad, India

- Explored inference optimization methods for large language models.
- Achieved a 3-4x speedup via optimization.

PROJECTS

Animatable 3D Avatar

Python, PyTorch, SMPL-X, Blender

April 2024

- Developed a motion transfer pipeline to animate 3D avatars using SMPL-X.
- Reconstructed 3D scenes and animated avatars within that.

CodeMix Generation [[GitHub](#)]

Python, PyTorch, NLP, mBART, mT5

April 2023

- Built a code-mixed translation system to generate Hinglish text by fine-tuning multilingual language models like mBART and MT5.

Text Conditioned 3D Scene Editing [[GitHub](#)]

Object Detection, Segmentation, Generative AI, NeRF

Nov 2023

- We tackled the problem of 3D scene editing, by editing a video consistently, using a novel approach, and recreating a scene from the video.

Customized xv6-OS [[GitHub](#)]

C, Linux System Calls, Schedulers

Oct 2021

- A modified version of MIT's xv6 Operating System, which adds several new scheduling algorithms- first come first serve, priority-based and multilevel feedback queue, along with some custom commands

Covid Central, Yavatmal [[GitHub](#)] [[Demo](#)]

Python, Dash, PostgreSQL

April 2021

- Worked with a group of four developers to create a platform to manage Covid-19 patients in hospitals.
- The system was provided to regional hospitals for no-charge.

Style Transfer for Headshot Portraits [[GitHub](#)]

Python, Image Processing, Dense Correspondence

Nov 2022

- Precise skills are crucial for compelling headshot editing, as errors can quickly degrade the quality of the photo.
- As a project of our DIP course, we developed a model which can transfer the style of the Example potrait to input image in a team of 4.

ACHIEVEMENTS

Dean's List - I	Monsoon 2023-24
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TECHNICAL SKILLS

Languages: Python, Java, C/C++, HTML/CSS
Frameworks: PyTorch, ONNX
Developer Tools: Git, Blender
Libraries: Open3D, OpenCV, pandas, NumPy, Matplotlib